

# STRAVYX

Australia's On-Demand Drone Services Network

## DJI Frontend Integration - Feature Map & Build Plan

Web (Next.js) + Mobile (iOS & Android) | Routes, Components, Primitives

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Source: docs/dji-frontend-integration.md. Companion to dji-integration-catalogue.md (what DJI offers) - this doc is how it lands in the frontend. Aligned to executive-summary.md and mvp-build-timeline.md.

Phasing rule: the 19 Jun MVP ships MANUAL equivalents (status timeline, file-list deliverables, manual upload, static admin lists). Everything DJI-powered here is Phase 1+ (2026-2027), designed to swap into the same screens without redesign. Routes/components are proposed conventions against the planned monorepo.

### 0. Assumptions & Conventions

- Monorepo: apps/{consumer,operator,admin,marketing}-web (Next.js 14 App Router); apps/{consumer,operator}-mobile (Expo) [P1].
- packages/ui, ui-native [P1], api-client, types + NEW: realtime (WS hooks), maps (Mapbox), media (players/viewers), dji-msdk (native bridge).
- Routes use App Router (app/.../page.tsx) matching Figma frame naming {app}/{feature}/{screen}.
- Visibility is server-enforced (NestJS visibility module); frontend renders only visibility-filtered payloads.
- DJI device MQTT is bridged server-side; the browser consumes it via packages/realtime WebSocket - never MQTT directly.

### 1. Shared Frontend Primitives (build once)

Most DJI features reduce to four cross-cutting primitives. Build these first.

| Primitive         | Package - Component                                  | Consumes                                          |
|-------------------|------------------------------------------------------|---------------------------------------------------|
| Real-time channel | realtime - useMissionStream, RealtimeProvider        | OSD telemetry, state/events, upload progress, HMS |
| Live map          | maps - LiveMap, RouteMap                             | Aircraft pins, site polygons, no-fly zones, docks |
| Video player      | media - LiveVideoPlayer (WebRTC/HLS)                 | Live camera feed (from RTMP/RTSP)                 |
| Geo / 3D viewers  | media - OrthoViewer, Model3DViewer, PointCloudViewer | FlightHub 2 Modeling API outputs                  |

Compose from planned packages/ui primitives: Button, Card, Badge, Modal, Table, Tabs, ProgressBar, Toast, Skeleton, EmptyState, StatusPill.

### 2. Consumer Web (apps/consumer-web)

| Feature              | Route                      | Components / primitives                                      | Phase           |
|----------------------|----------------------------|--------------------------------------------------------------|-----------------|
| Live tracker (C-05)  | missions/[id]              | MissionTracker -> LiveMap + StatusTimeline; useMissionStream | MVP->P1 live    |
| Watch live feed      | missions/[id]/live         | LiveFeedPanel -> LiveVideoPlayer; Modal/Tabs                 | P1              |
| Deliverables (C-06)  | missions/[id]/deliverables | DeliverableGallery, OrthoViewer, Model3DViewer               | MVP->P1 viewers |
| Auto status          | (all mission screens)      | MissionStatusProvider; StatusPill/Toast                      | P1              |
| Deliverable QR share | missions/[id]/deliverables | ShareModel; Modal/Button + QR util                           | P1              |

### 3. Mobile Operator Web (apps/operator-web)

| Feature                    | Route                | Components / primitives                             | Phase   |
|----------------------------|----------------------|-----------------------------------------------------|---------|
| Device binding (O-01)      | onboarding/devices   | DeviceBindingPanel; Card/Button/Modal               | P1      |
| Assigned route (O-03)      | missions/[id]        | RouteMap (Mapbox KMZ)                               | MVP->P1 |
| Auto upload status (O-09)  | missions/[id]/upload | UploadProgress (event-driven); ProgressBar/Table    | MVP->P1 |
| Flight telemetry + HMS     | missions/[id]        | FlightStatusPanel, HmsAlertBanner; useMissionStream | P1      |
| Verified capability (O-02) | profile/capabilities | CapabilityForm (prefill from topology)              | MVP->P1 |

#### 4. Admin Web (apps/admin-web)

| Feature                     | Route           | Components / primitives                                | Phase   |
|-----------------------------|-----------------|--------------------------------------------------------|---------|
| Live fleet map (A-02)       | fleet           | FleetMap; LiveMap + useFleetStream                     | MVP->P1 |
| Health / HMS dashboard      | health          | HealthDashboard, HmsFeed; Table/Badge/Toast            | P1      |
| Dock control panel          | docks/[id]      | DockControlPanel (open/close, start/stop, forced land) | P1-2    |
| Situational awareness (TSA) | fleet (overlay) | TsaOverlay; LiveMap layer                              | P1+     |
| Livestream monitoring       | missions/[id]   | MultiStreamViewer; Tabs + LiveVideoPlayer              | P1      |

Dock control is safety-critical: every action uses a Modal confirm + writes to the audit log (M-07); controls disabled unless dock/aircraft state reports ready.

#### 5. Marketing Web (apps/marketing-web)

- technology page: LiveTrackingDemo (canned) + Sample3D (Model3DViewer) as DJI credibility content. [P1]
- operators page: OperatorSignupForm (K-04). [MVP] No live SDK wiring - embeds sample deliverables only.

#### 6. Mobile Apps - iOS & Android (Phase 1, post-GA)

##### 6.1 Architecture decision

MSDK v5 is native (Android AAR / iOS framework) and cannot run in Expo Go. Approach: React Native + Expo with a custom dev client (config plugin, prebuild/bare), keeping JS velocity + shared packages with web.

- packages/dji-msdk - native module wrapping MSDK, exposed to JS: registerApp, connectProduct, startWaypointMission(kmz), startLiveStream(url), onTelemetry(cb), uploadMedia().
- Android: Kotlin, Gradle com.dji:dji-sdk-v5-aircraft, -provided (compileOnly), -networkImp (runtime).
- iOS: Swift module linking DJI MSDK v5 via CocoaPods/SPM.
- Two apps share packages/ui-native: consumer-mobile (booking + tracker + deliverables) and operator-mobile (the flight app - real MSDK consumer).

##### 6.2 Mobile feature map

| Feature                | App               | dji-msdk / native call            | Notes                          |
|------------------------|-------------------|-----------------------------------|--------------------------------|
| Register + connect     | operator          | registerApp, connectProduct       | App key; RC link state         |
| Fly assigned mission   | operator          | startWaypointMission(kmz)         | One-tap KMZ; auto status       |
| Pre-flight gate (O-07) | operator          | reads FC/HMS state                | Blocks READY->AIRBORNE (D-04)  |
| Live telemetry         | operator/consumer | onTelemetry(cb)                   | Same UX as web tracker         |
| Live stream            | operator          | startLiveStream(url)              | Feeds consumer watch-live      |
| Media + auto upload    | operator          | IMediaManager + uploadMedia()->S3 | Replaces manual; fires Layer 2 |
| Camera / gimbal specs  | operator          | camera/gimbal APIs                | Per-category capture           |
| Tracker + deliverables | consumer          | none (realtime + API)             | Shared with web logic          |

##### 6.3 Platform specifics

| Concern         | Android                                | iOS                                    |
|-----------------|----------------------------------------|----------------------------------------|
| SDK integration | Gradle AAR (com.dji:dji-sdk-v5-*)      | CocoaPods/SPM MSDK v5 framework        |
| RC connection   | USB accessory intent + permission      | MFi external accessory entitlement     |
| Background      | Foreground service (telemetry/upload)  | Background modes (accessory, audio)    |
| Min OS          | Android 8+                             | iOS 14+                                |
| Store review    | Play: accessory + location disclosures | App Store: MFi + background + location |

|          |                                  |                               |
|----------|----------------------------------|-------------------------------|
| Build/CI | Expo prebuild + EAS (dev client) | EAS; Apple provisioning + MFi |
|----------|----------------------------------|-------------------------------|

## 7. Build Plan

### 7.1 Web - DJI frontend (Phase 1, post-GA)

| Sprint | Focus                       | Deliverables                                                                       |
|--------|-----------------------------|------------------------------------------------------------------------------------|
| W-P1.1 | Realtime foundation         | packages/realtime (WS + useMissionStream); server WS bridge; maps LiveMap skeleton |
| W-P1.2 | Consumer live tracker       | C-05 timeline -> LiveMap live pin; auto status provider                            |
| W-P1.3 | Media return + deliverables | UploadProgress (operator); DeliverableGallery + OrthoViewer (consumer)             |
| W-P1.4 | Live video                  | media LiveVideoPlayer; consumer watch-live; admin MultiStreamViewer                |
| W-P1.5 | 3D deliverables             | Model3DViewer / PointCloudViewer wired to FlightHub 2; QR share                    |
| W-P1.6 | Admin fleet + health        | FleetMap, HealthDashboard, HmsFeed; operator device binding                        |
| W-P1.7 | Dock control + hardening    | DockControlPanel (audited, confirm modals); TSA overlay; a11y + perf               |

### 7.2 Mobile - iOS & Android (Phase 1 programme)

| Sprint | Focus              | Deliverables                                                                 |
|--------|--------------------|------------------------------------------------------------------------------|
| M-P1.0 | Foundation         | Expo monorepo apps; ui-native; custom dev client; auth + api-client reuse    |
| M-P1.1 | dji-msdk (Android) | Kotlin: registerApp/connectProduct/onTelemetry; RC USB; connect screen       |
| M-P1.2 | dji-msdk (iOS)     | Swift parity; MFi entitlement; connect screen on iOS                         |
| M-P1.3 | Fly mission        | startWaypointMission(kmz); PreflightChecklist gate; FlightHud telemetry      |
| M-P1.4 | Media + upload     | IMediaManager review + uploadMedia()->S3; auto Layer 2 trigger               |
| M-P1.5 | Live stream        | startLiveStream; wire to consumer watch-live                                 |
| M-P1.6 | Consumer app       | Booking reuse, MissionTracker, DeliverableGallery (shared logic)             |
| M-P1.7 | Store readiness    | EAS builds; Play + App Store listings; disclosures; TestFlight/internal beta |

### 7.3 Sequencing & dependencies

Hard dependencies: the NestJS Cloud API bridge (server subscribes to DJI MQTT, re-publishes visibility-filtered data over WS/REST) gates ALL web realtime features; the dji-msdk native bridge gates all operator-mobile flight features.

Cut lines (if Phase 1 slips): ship web live tracker + auto upload first (highest UX/ops value); defer 3D viewers, dock control, and the full operator-mobile flight app.

## 8. Definition of Done (DJI Frontend, Phase 1)

1. Consumer sees a live moving aircraft pin + auto-updating status on the tracker.
2. Operator media auto-uploads with visible progress; Layer 2 job fires without manual steps.
3. Consumer views processed 2D/3D deliverables in-portal (viewer, not just download).
4. Admin sees a live fleet map + HMS health; dock controls confirm-gated and audited.
5. operator-mobile can connect a drone and fly a Stravix KMZ mission on iOS and Android.
6. All DJI data remains visibility-filtered - no customer total / Layer 2 fee exposed to operators.

Markdown source: docs/dji-frontend-integration.md. Related: dji-integration-catalogue.md, executive-summary.md, mvp-build-timeline.md. Routes/components are proposed - reconcile with Figma at build.